



UNIVERSITÀ DI PISA

MSc in Data Science and Business Informatics

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BUSINESS PROCESS MODELLING · PROJECT REPORT

# P29: Thesis

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# 1 Scenario and approach

Consider the (suitably simplified) scenario of the preparation of a degree thesis by a student.

The student contacts a professor, expressing interest in writing a thesis in the professor's area of expertise. The professor proposes a set of possible topics, and the student either picks one or informs the professor that they are no longer interested in the thesis. If the professor has been chosen as supervisor, they send the student some material to study.

After studying the topic, the student starts developing the thesis, periodically exchanging documents, chapter drafts and software prototypes with the supervisor, and scheduling meetings to discuss the work in depth. Each time the student sends an object, the supervisor replies by annotating the object with comments. Each time a meeting must be scheduled, the student proposes a date, which the supervisor may confirm or reject by proposing an alternative date, and so on until a date suits both.

When the thesis is complete, the student sends it to a counter-examiner for evaluation. The counter-examiner may decide to schedule an appointment with the student for a direct presentation of the work, or deem such a meeting unnecessary. The date of the meeting, if any, is set as described above.

The process ends when the supervisor and the counter-examiner send their evaluations to the committee which, after examining the student and reading the evaluations, meets to decide the graduation grade and communicates it to the student.

Design suitable processes that faithfully mirror the scenario described above and that interact correctly.

The assignment above describes the thesis-preparation process with few gaps, and we close each one with a decision of our own.

- **Exchanging objects and scheduling meetings at once** is never addressed, and we keep the two exclusive.
- **No one is appointed to call the thesis complete**, so we give that verdict to the supervisor and leave the student to ask for it.
- **Nothing prompts the two evaluations**, and we fix one for each: the supervisor sends on approving the thesis, the counter-examiner once the presentation is settled or waived.
- **The examination is not ordered against the two evaluations**, and we have the committee convoke the student only after both have arrived.

We model the scenario in BPMN, translate each diagram into a workflow net, and run the same checks twice: on each pool alone (??), then on the four composed (??). The two answers diverge, and that divergence is the report's subject: every pool is sound by itself, while the collaboration is not. [Table 1](#) maps each requirement of the assignment to the place that discharges it, and [Table 2](#) lists the tools.

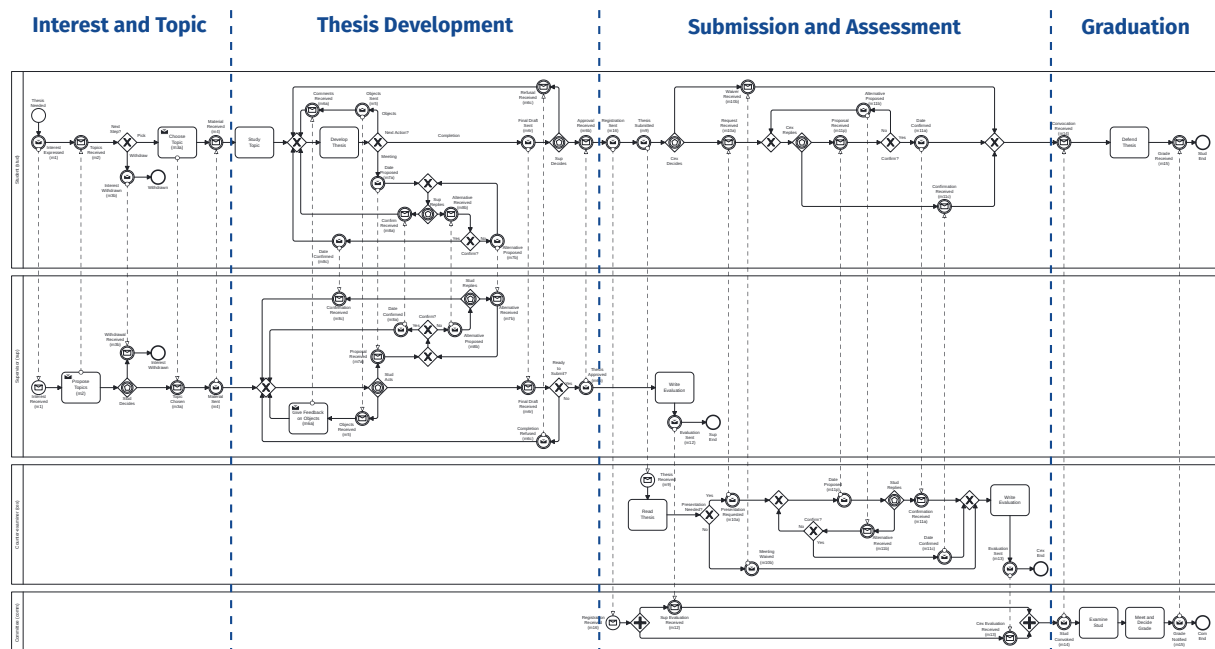
| What the assignment asks                                                             | Where                                              |
|--------------------------------------------------------------------------------------|----------------------------------------------------|
| the design choices                                                                   | <a href="#">Sections 1</a> and <a href="#">2</a>   |
| processes that faithfully mirror the scenario                                        | <a href="#">Section 2</a>                          |
| processes that interact correctly                                                    | <a href="#">Section 2.2</a> and <a href="#">??</a> |
| models drawn at an abstract level, then transformed into workflow nets               | <a href="#">Sections 2</a> and <a href="#">2.3</a> |
| how the transformation was obtained                                                  | <a href="#">Section 2.3</a>                        |
| invariants, S-components, free-choice, well-handled (well-structuredness), safeness  | ????                                               |
| an estimate of the reachability-graph size in vertices and arcs                      | ????                                               |
| soundness of each participant's orchestration                                        | ??                                                 |
| soundness of the whole collaboration                                                 | ??                                                 |
| if no sound overall process is possible: the reason, with relaxed and weak soundness | ??                                                 |
| diagrams and nets in a standard interchange format                                   | bpmn/, pnml/                                       |
| analysis output and reachability-graph images                                        | pnml/analysis/,<br>pnml/img/                       |

**Table 1:** Every requirement of the assignment against the place that discharges it.

| Step and tool                                                                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Drawing</b> ( <a href="#">bpmn.io</a> )<br>Visual Studio Code extension, editing native BPMN 2.0 XML.                                                                                                                                                                                  |
| <b>2. Validation</b> ( <a href="#">bpmnlint</a> 11.12.1)<br><a href="#">bpmnlint:all</a> in <code>scripts/lint/.bpmnlintrc</code> , run by <code>npm --prefix scripts/lint run lint</code> .                                                                                                 |
| <b>3. Transformation</b> ( <a href="#">bpmn2petrinet</a> )<br>Browser tool, cloned from <a href="#">BenjaNapo/bpmn-to-petri</a> and driven headless by <code>scripts/03_to_pnml.py</code> with its <code>withCollapsedXor</code> setting on.                                                 |
| <b>4. Analysis</b> ( <a href="#">pm4py</a> 2.7.23.6)<br><code>pm4py</code> 's reimplementaion of Woflan reached as a library by <code>scripts/04_pm4py_report.py</code> . WoPeD spreads the same figures across panels, so collecting them would take a script reading numbers off a screen. |
| <b>5. Rendering</b> ( <code>pm4py</code> and <a href="#">WoPeD</a> )<br><code>scripts/04_pm4py_report.py</code> draws each net and its reachability graph, WoPeD the same nets in a second layout, all under <code>pnml/img/</code> .                                                        |

**Table 2:** The pipeline from diagram to analysis.

## 2 From BPMN modelling to Petri nets



**Figure 1:** The collaboration (bpmn/collaboration.bpmn), cut into the four phases of [Section 2.2](#).

**Figure 1** is the whole model: one row per participant, four orchestrations in all ([Section 2.1](#)), running together through the four phases of the collaboration ([Section 2.2](#)). No wait in it carries a deadline, since we assume every participant answers in the end, so the model has no timer and no branch for an answer that never comes.

### 2.1 Orchestrations

In each orchestration, and in the collaboration, a joint act takes a task only where the messages alone do not show it happened. When both participants send or receive the message that settles it, as with the acceptance of a meeting date, it takes none. If only one side sends, as with the convocation to the examination, nothing shows that the student attended, and the act takes a task on each side. [Table 3](#) counts the elements of each orchestration.

|                           | Student | Supervisor | Counter-examiner | Committee | Total  |
|---------------------------|---------|------------|------------------|-----------|--------|
| start / end events        | 1 / 2   | 1 / 2      | 1 / 1            | 1 / 1     | 4 / 6  |
| tasks (send tasks)        | 4 (1)   | 3 (2)      | 2 (0)            | 2 (0)     | 11 (3) |
| intermediate catch events | 13      | 7          | 2                | 2         | 24     |
| intermediate throw events | 11      | 6          | 5                | 2         | 24     |
| event-based gateways      | 4       | 3          | 1                | 0         | 8      |
| exclusive gateways        | 8       | 4          | 4                | 0         | 16     |
| parallel gateways         | 0       | 0          | 0                | 2         | 2      |
| nodes                     | 43      | 26         | 16               | 10        | 95     |
| sequence flows            | 50      | 30         | 18               | 10        | 108    |

**Table 3:** Element counts for the four orchestrations.

**Student** (`bpmn/student.bpmn`). Each round of the development loop opens on Develop Thesis and reaches the XOR-split Next Action?, which offers three moves towards the supervisor: send an object, propose a meeting date, or send the final draft and ask whether the work is done. Keeping them exclusive rules out the runs where the supervisor is annotating a draft while a date is still open, and it leaves the pool an S-net.

Once the supervisor approves, the student registers for the graduation call and only then sends the thesis to the counter-examiner. The student waits to hear whether the counter-examiner wants a presentation, confirms or counters the date they propose, and ends on the convocation, Defend Thesis and the grade.

**Supervisor** (`bpmn/supervisor.bpmn`). The student's withdrawal closes this orchestration at once. Otherwise it mirrors the student's, with one wait for the three branches a round can take and another after the supervisor counter-proposes a date. It ends when the supervisor writes the evaluation and sends it to the committee.

**Counter-examiner** (`bpmn/counter-examiner.bpmn`). Read Thesis precedes the XOR-split Presentation Needed?. The counter-examiner either waives the meeting or asks for one and proposes a date, which the student confirms or counters until both agree. The branches merge, and Write Evaluation follows, so whatever the presentation settles reaches the evaluation.

**Committee** (`bpmn/committee.bpmn`). Nothing orders the supervisor's evaluation against the counter-examiner's, and the assignment makes the examination and the reading two prerequisites of the same meeting, so the orchestration waits for the two evaluations in parallel and joins before convoking the student. That is the one AND in the model.

## 2.2 Collaboration

Twenty-seven message flows wire the four pools in the collaboration ([Table 4](#)). The student's pool is the axis: it touches twenty-five, missing only the two evaluations that reach the committee. The student also instantiates all three of the others, one message each. Because that pool runs through all of it, the collaboration reads as four phases, drawn over [Figure 1](#) as bands rather than as part of the diagram.

**Interest and Topic.** The student contacts the supervisor, who replies with a list of topics and from then on only reacts, since nothing in that orchestration starts on its own. The student picks a topic or withdraws, the only point at which the case can be abandoned.

**Thesis Development.** The core of the model is a loop of request and response. The student and the supervisor step together, one message and its answer per round, and the loop ends on the supervisor's verdict, so both leave development together.

**Submission and Assessment.** The registration instantiates the committee, and the thesis the counter-examiner. The supervisor's evaluation could have started the committee instead; the registration keeps the student the only one who starts another participant.

**Graduation.** The committee convokes the student and the two hold the examination, a task in each pool. The committee then decides the grade and sends it, closing its own orchestration and the student's.

| #  | Msg                    | From | To   | Content                            | Phase                     | In the assignment |
|----|------------------------|------|------|------------------------------------|---------------------------|-------------------|
| 1  | <i>m<sub>1</sub></i>   | stud | sup  | expression of interest             | Interest and Topic        | ✓                 |
| 2  | <i>m<sub>2</sub></i>   | sup  | stud | topic list                         | Interest and Topic        | ✓                 |
| 3  | <i>m<sub>3a</sub></i>  | stud | sup  | topic choice                       | Interest and Topic        | ✓                 |
| 4  | <i>m<sub>3b</sub></i>  | stud | sup  | withdrawal                         | Interest and Topic        | ✓                 |
| 5  | <i>m<sub>4</sub></i>   | sup  | stud | study material                     | Interest and Topic        | ✓                 |
| 6  | <i>m<sub>5</sub></i>   | stud | sup  | draft / object                     | Thesis Development        | ✓                 |
| 7  | <i>m<sub>6a</sub></i>  | sup  | stud | comments, keep going               | Thesis Development        | ✓                 |
| 8  | <i>m<sub>6b</sub></i>  | sup  | stud | ready to submit                    | Thesis Development        |                   |
| 9  | <i>m<sub>6c</sub></i>  | sup  | stud | not yet, keep going                | Thesis Development        |                   |
| 10 | <i>m<sub>6r</sub></i>  | stud | sup  | final draft and completion request | Thesis Development        |                   |
| 11 | <i>m<sub>7a</sub></i>  | stud | sup  | meeting-date proposal              | Thesis Development        | ✓                 |
| 12 | <i>m<sub>7b</sub></i>  | stud | sup  | alternative date                   | Thesis Development        | ✓                 |
| 13 | <i>m<sub>8a</sub></i>  | sup  | stud | date confirmed                     | Thesis Development        | ✓                 |
| 14 | <i>m<sub>8b</sub></i>  | sup  | stud | counter-proposal                   | Thesis Development        | ✓                 |
| 15 | <i>m<sub>8c</sub></i>  | stud | sup  | counter-date accepted              | Thesis Development        |                   |
| 16 | <i>m<sub>9</sub></i>   | stud | cex  | completed thesis                   | Submission and Assessment | ✓                 |
| 17 | <i>m<sub>10a</sub></i> | cex  | stud | presentation requested             | Submission and Assessment | ✓                 |
| 18 | <i>m<sub>10b</sub></i> | cex  | stud | no meeting needed                  | Submission and Assessment |                   |
| 19 | <i>m<sub>11p</sub></i> | cex  | stud | presentation-date proposal         | Submission and Assessment | ✓                 |
| 20 | <i>m<sub>11a</sub></i> | stud | cex  | date confirmed                     | Submission and Assessment | ✓                 |
| 21 | <i>m<sub>11b</sub></i> | stud | cex  | counter-proposal                   | Submission and Assessment | ✓                 |
| 22 | <i>m<sub>11c</sub></i> | cex  | stud | counter-date accepted              | Submission and Assessment |                   |
| 23 | <i>m<sub>12</sub></i>  | sup  | comm | supervisor evaluation              | Submission and Assessment | ✓                 |
| 24 | <i>m<sub>13</sub></i>  | cex  | comm | counter-examiner evaluation        | Submission and Assessment | ✓                 |
| 25 | <i>m<sub>16</sub></i>  | stud | comm | graduation registration            | Submission and Assessment |                   |
| 26 | <i>m<sub>14</sub></i>  | comm | stud | exam convocation                   | Graduation                |                   |
| 27 | <i>m<sub>15</sub></i>  | comm | stud | grade                              | Graduation                | ✓                 |

**Table 4:** Message inventory: twenty-seven flows in the delivered model, each with exactly one sender and one receiver, so the four modules are structurally compatible and compose. **stud** is the student, **sup** the supervisor, **cex** the counter-examiner and **comm** the committee. The last column marks whether the assignment names a message; the unmarked ones are ours.

## 2.3 Deriving the nets

`bpmnlint` reports nothing on the six diagrams, and `bpmn2petrinet` then translates them with `withCollapsedXor` on. Every exclusive and event-based gateway collapses into one place, and its branches compete on that place; every other node becomes a transition. A sequence flow becomes a place between two transitions, or a plain arc where a collapsed gateway sits at one end. We drive the converter headless from `scripts/03_to_pnml.py`, so no step between diagram and net is manual.<sup>1</sup>

The tool builds the workflow net itself. It puts a source place in front of the start event and marks that place with one token, and it merges the end events of a process into a single sink, so the six pool ends become four. Every message flow gets one place, shared by the sending and the receiving transition and reused on every round. One transition then closes the collaboration, collecting the four pool sinks into a final place.

`scripts/01_extract_pools.py` cuts each participant out of `bpmn/collaboration.bpmn` into a diagram of its own, so the analysis has both the collaboration and the four orchestrations from the same model. Read alone, an orchestration carries a source of its own and no message places; inside the collaboration the message places return, and the student's is the only source.

<sup>1</sup>The converter departs from the PNML schema in two places: it writes `<position>` where the schema asks for `<offset>`, and it leaves the `<net>` without the `id` and `type` the standard requires. We repair both. Neither changes the analysis, since `pm4py` reads the file either way; they are there for the tools that validate before reading and return nothing on a file that fails.

**Table 5** reconciles the two readings, and its transitions follow from **Table 3** up to one correction: a flow that runs from one collapsed gateway into another would leave two places adjacent, so it becomes a transition of its own. There are two such flows in the student, two in the supervisor, one in the counter-examiner and none in the committee, which gives  $43 - 12 + 2 = 33$  from 43 nodes and 12 collapsed gateways, and the same arithmetic for the rest. The places follow the same way: one for each collapsed gateway, one for each flow left with a transition at both ends, and the source and the sink, so  $12 + 11 + 2 = 25$  for the student.

|                                             | Source                     | Places | Transitions | Arcs |
|---------------------------------------------|----------------------------|--------|-------------|------|
| Student                                     | pnml/student.pnml          | 25     | 33          | 66   |
| Supervisor                                  | pnml/supervisor.pnml       | 16     | 21          | 42   |
| Counter-examiner                            | pnml/counter-examiner.pnml | 10     | 12          | 24   |
| Committee                                   | pnml/committee.pnml        | 12     | 10          | 22   |
| the four pools                              |                            | 63     | 76          | 154  |
| sources the message-started pools lose      |                            | -3     | 0           | -3   |
| the twenty-seven message places             |                            | +27    | 0           | +54  |
| closure, the final place and its transition |                            | +1     | +1          | +5   |
| collaboration                               | pnml/collaboration.pnml    | 88     | 77          | 210  |

**Table 5:** The final nets. The three middle rows reconcile the system with the four pools.